Oil Price Shocks and the UK Economy: Analysis of the Consequences of a Global Economy under recent events.

1. Introduction

1.1 Background

The impact of global political events on the price of oil and, subsequently, the global economy, have been analysed extensively since the 1970s global recession following the Yom Kippur War in 1973 which led to the increase in the real price of oil due to an OPEC embargo (Darby 1982). Theoretically, it constitutes the third factor in the aggregate production function, capital. Thus, when the oil prices increased in the early 1970s, the aggregate supply curve shifted accordingly producing lower output and a higher price level in global economies. Though works by Michael Darby (1982) and James Tobin (1980) contest this theoretical relationship, the significance of our economic dependence on oil was clearly illustrated with the subsequent decline of real income and rise of unemployment in the UK. With geo-political tensions in the middle east rising once again and the red sea restricted, the threat oil price volatility on our economy has significant weight. In this piece, I will analyse the UK's economic sensitivity to global oil price volatility in the context of significant global events and explore the UK's macroeconomic indicators' reaction to global oil price shocks through time series analysis of the price of crude oil on key macroeconomic indicators.

1.2 Literature review

The relationship between crude oil prices and macroeconomic indicators has been investigated closely, nonetheless, the evolving dynamics of globalised and interdependent economies warrants constant re-examination of this relationship. The instability of oil prices over time (Edelstein and Kilian 2009, Kilian 2008) raises difficult challenges in time series modelling involving oil prices as outdated models bear less relevance in the modern day. Moreover, there is significant evidence (Peersman and Van Robays, 2009; Kilian 2009) which illustrates how; as the nature of oil demand and supply shocks evolves, so too does the relationship of the real price of oil to the economy.

The seminal analysis by Hamilton (Hamilton, 1983) on recessions post World War II identifies a significant, lagged relationship between shocks in the price of crude oil and all but one recession in the US, as of the 1983 publication. Moreover, the author suggests that the oil industry, experiencing erosion of the real price of oil due to inflation, opportunistically made up for this loss during significant global events, exogenous in context to the economy, namely the nationalisation of Iranian assets, the Suez Crisis and a general decline in energy reserves. This piece looks to expand and

investigate this premise in the context of recent events' impacts, post September 2022, on the UK economy. Specifically, OPEC's non-negligible role in causing oil price fluctuations with announcements and restrictions to oil production has been found to have a significant relationship with macroeconomic aggregates (Lorusso and Pieroni, 2018) and parallels can be made between the G7's sanctions on Russian oil and the 1973-1974 OPEC embargo which had severe impacts on the US economy and the UK's economy by consequence. Though it should be noted that recent VAR and GARCH analysis by Wei Zhao (20203) has shown that, out all the G7 countries, the UK's stock market was least effected during the Russia Ukraine war due to the UK's indigenous crude oil production and non-proximity to war zones, in contrast to other G7 countries.

As Escribano (2017) discusses, an academic consensus on the "nature of the oil-stock nexus" is contested. Previous scholar's analysis of this relationship clearly shows variability in results with developed countries (Kilian and Park, 2009, Nandha and Faff, 2008), emerging economies (Hammoudeh and Choi, 2007) (Hammoudeh and Li, 2005) and oil importing/exporting economies yielding distinct relationships with price shocks in oil. Work by Kilian and Park (2009) illustrates the change in stock prices is also dependent on whether the changes in crude oil prices was driven by a supply or demand shock. As such, analysis from this piece mustn't be used as a generalisation for this complex relationship but rather an unravelling of the UK's economic dependence on oil prices.

2. Methodology

2.1 Data sources

Data for this study was taken from the Wall Street Journal, the Bank of England and the Energy Information Administration, based on the daily prices of the UK's FTSE, the Sterling/Dollar exchange rate and Brent crude oil prices respectively. The data spans from the 30th of August 2021 to the 19th of November 2023 and was interpolated to account for missing figures and converted to weekly data as to capture significant trends and exclude daily, 'noise-driven' market fluctuations that do not adequately reflect market movements (Hamilton 1994). Moreover, this approach can capture the cumulative nature of reactions to significant global events in these micro economies, both leading up to the, often scheduled, event and post-event (Schwert, 1989). Brent crude oil prices were not adjusted for seasonality as to capture deterministic changes in the analysis.

2.2 ADF test

The ARIMA (Autoregressive Integrated Moving Average) and VAR (Vector Autoregressive) models both rely on the assumption that data involved be stationary. As such, the ADF (Augmented Dickey-Fuller) test was required to examine the presence of unit roots in the data. Following ADF testing, all

data sets proved not to reject the null hypothesis, that the series has a unit root. By taking the log and differencing each series and then performing the test again, produced p-values indicated stationarity was achieved.

2.3 VAR modelling

VAR (Vector Autoregression) model is a common model used in time series analysis to investigate the dynamics of several endogenous variables. In this piece, two bivariate VAR models were included. The first examines the relationship between the Sterling/Dollar exchange rate and brent crude oil prices and the second examines the relationship between the FTSE100 and brent crude oil prices.

2.3.1 VAR model 1:

$$P_t^{Oil} = a_1 + b_{11}P_{t-1}^{Oil} + b_{12}R_{t-1}^{usdgbp} + \varepsilon_t^{Oil}$$

$$R_t^{UsdGbp} = a_2 + b_{21}P_{t-1}^{Oil} + b_{22}R_{t-1}^{usdgbp} + \varepsilon_t^{UsdGbp}$$

$$\begin{bmatrix} P_t^{Oil} \\ R_t^{UsdGbp} \end{bmatrix} = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} + \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} \begin{bmatrix} P_{t-1}^{Oil} \\ R_{t-1}^{UsdGbp} \end{bmatrix} + \begin{bmatrix} \varepsilon_t^{Oil} \\ \varepsilon_t^{UsdGbp} \end{bmatrix}$$

2.3.2 VAR model 2:

$$P_t^{Oil} = a_1 + \sum_{m=1}^{m=3} b_{11}P_{t-1}^{Oil} + \sum_{m=1}^{m=3} b_{12}R_{t-1}^{usdgbp} + \varepsilon_t^{Oil}$$

$$F_t^{FTSE} = a_2 + \sum_{m=1}^{m=3} b_{21}P_{t-1}^{Oil} + \sum_{m=1}^{m=3} b_{22}F_{t-1}^{FTSE} + \varepsilon_t^{FTSE}$$

$$\begin{bmatrix} P_t^{Oil} \\ F_t^{FTSE} \end{bmatrix} = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} + \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} \begin{bmatrix} P_{t-1}^{Oil} \\ F_{t-1}^{FTSE} \end{bmatrix} + \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} \begin{bmatrix} P_{t-2}^{Oil} \\ F_{t-2}^{FTSE} \end{bmatrix} + \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} \begin{bmatrix} P_{t-3}^{Oil} \\ F_{t-3}^{FTSE} \end{bmatrix} + \begin{bmatrix} \varepsilon_t^{Oil} \\ \varepsilon_t^{FTSE} \end{bmatrix}$$

2.4 Test for unknown breaks

Table 1. *Unknown break test*

Test	Test statistic	1% Critical Value	5% Critical Value	10% Critical Value
F(1|0)	4.25	7.68	5.74	`4.91
F(2|1)	22.14	8.42	6.47	5.70
F(3|2)	24.34	8.86	7.01	6.14
F(4|3)	10139.37	9.34	7.42	6.45
F(5|4)	22591.58	9.59	7.64	6.74

Note: Rounded to 2D.P

As to assist in determining the break points or global events of interest, a sequential test for unknown break points (Bai and perron, 2003) in the Brent crude oil prices was constructed as presented in table 1.

2.5 Break points and descriptions

From this information, and considering recent and relevant global events, break points in the following weeks were tested individually and in ARIMA models.

2.5.1 2022 Week 49 (break 1):

The G7+ price cap on Russian exported oil was put into effect at $60 a barrel. Despite discounted Russian oil, especially to non G7+ countries such as India, raising oil exports in November, we saw a $3bn revenue loss in December for Russia. The G7+ price cap may well have caused a supply shock in the global market, though tempered by an OPEC+ production drop of 870 kb/d in anticipation (IEA 2022).

2.5.2 2023 Week 10 (break 2):

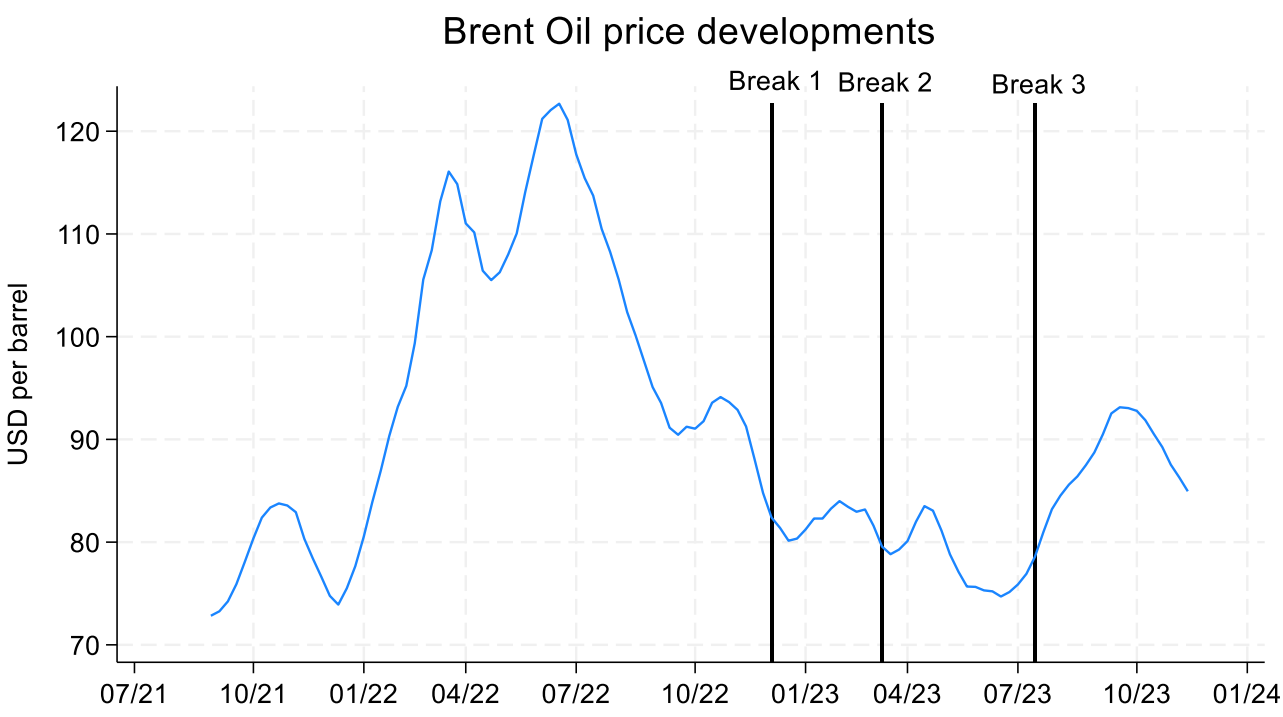
The rise of a Russian “dark fleet” exporting sanctioned oil combined with a ‘supply overhang’ as stocks build up to an 18-month high created stagnation in the market. This was then broken by the Silicon Valley Bank collapsing, leading to doubt in the financial sector and a drop in demand for Brent futures (IEA 2023a)

2.5.3 2023 Week 28 (break 3):

OPEC+ met for scheduled oil output policy and announced it will reduce production targets further in 2024 by a total of 1.4 million bpd. OPEC+ sources over 40% of worldwide oil, however, a growing China that, at the time, was steadfast in recovering from the pandemic may have lessened the impact of the policy on the global economy (IEA 2023b).

3. Results and analysis

Figure 1.



Notes: The lines mark the aforementioned significant global events and chosen breakpoints. Break 1: 2022w49 Break 2: 2023w10 Break 3: 2023w28
Source: Federal Reserve Bank of St. Louis

3.1 ADF tests

The methodology of this piece utilises both ARIMA and VAR models. In order to avoid the circumstance of pseudo-vector autoregression, the augmented dickey fuller test (ADF) was used to test the data. By taking the log and the first difference each variable, in that order, stationarity was achieved. Table 2 illustrates that all variables had unit roots prior to differencing and logging and, following retesting, the test yielded p-values of 0.

Table 2. *ADF Test*

Variables:	t-statistic:	p-values:
FTSE100	-2.666	0.080*
Sterling/Dollar	-1.802	0.379
Brent crude oil	-1.817	0.372
Logged and first differenced		
FTSE100	-8.669	0.000***
Sterling/Dollar	-8.190	0.000***
Brent crude oil	-9.717	0.000***

*Note: * Indicates statistical significance at the 10% level. ** Indicates significance at the 5% level and *** at the 1% level*
Results are rounded to 3D.P

3.2 VAR lag selection

In order to determine the lag selection, VAR lag selection tests, specifically LL, LR, FPE, AIC and HQIC tests were used on both VAR models. These results are depicted in tables 3 and 4. But as Zhao (2023) points out, these tests maybe misleading given stock and currency exchange markets respond swiftly to changes in crude oil prices. Therefore, the comparative results for Eigenvalue stability conditions of VAR models were also used to determine the lag selection and the VAR lag selection tests were not strictly relied upon.

Table 3. *VAR model 1 identification*

LAG	LL	LR	DF	P-VALUE	FPE	AIC	HQIC	SBIC
0	539.333				2.10E-07	-9.6816	-9.6618*	-9.6328*
1	544.705	10.745*	4	0.030	2.10E-07*	-9.7064*	-9.6469	-9.5599
2	545.191	0.972	4	0.914	2.20E-07	-9.6430	-9.5440	-9.3989
3	547.267	4.150	4	0.386	2.30E-07	-9.6084	-9.4697	-9.2666
4	548.714	2.895	4	0.575	2.40E-07	-9.5624	-9.3841	-9.1230

Note: * *Indicates the recommended lag length by the respective test*

***Table 4**. VAR model 2 identification*

Lag	*LL*	*LR*	*df*	*p*	*FPE*	*AIC*	*HQIC*	*SBIC*
0	501.378			-	4.2e-07	-8.9978	-8.9780	-8.9489*
1	507.078	11.400	4	0.022	4.1e-07	-9.0284	-8.9690	-8.8819
2	516.051	17.946*	4	0.001	3.8e-07*	-9.1180*	-9.0190*	-8.8739
3	517.571	3.040	4	0.551	3.9e-07	-9.0733	-8.9347	-8.7316
4	520.231	5.319	4	0.256	4.0e-07	-9.0492	-8.8709	-8.6098

Note: * *Indicates the recommended lag length by the respective test*

3.3 VAR Granger-causality tests

Granger-causality tests are conducted in order to examine the weight of lagged variables in predicting other variables. Whilst granger-causality tests utilising OLS regression only produced statistically significant results for the regression of lagged variables of Brent crude oil prices onto the FTSE100, VAR granger causality tests provided more nuanced results, as shown in table 3. The differing results maybe as a result of the OLS regression's inability to capture the dynamic interrelationships of the variables, including feedback loops and other simultaneous interactions (Enders, 2015) or, more simply, as a result of VAR analysis capturing the optimal lag structure.

***Table 5**. Granger-causality with VAR*

Direction of Causality	**Chi-Squared**	**Lags**	**p-value**
Brent Oil Prices → FTSE100	13.46	3	0.04**
FTSE100 → Brent Oil Prices	1.86	3	0.05**
Brent Oil Prices → Sterling/Dollar exchange	0.08	1	0.76
Sterling/Dollar exchange → Brent Oil Prices	2.93	1	0.09*

*Note: * indicates a significance level below 10% and ** below 5%. Results rounded to 2D.P*

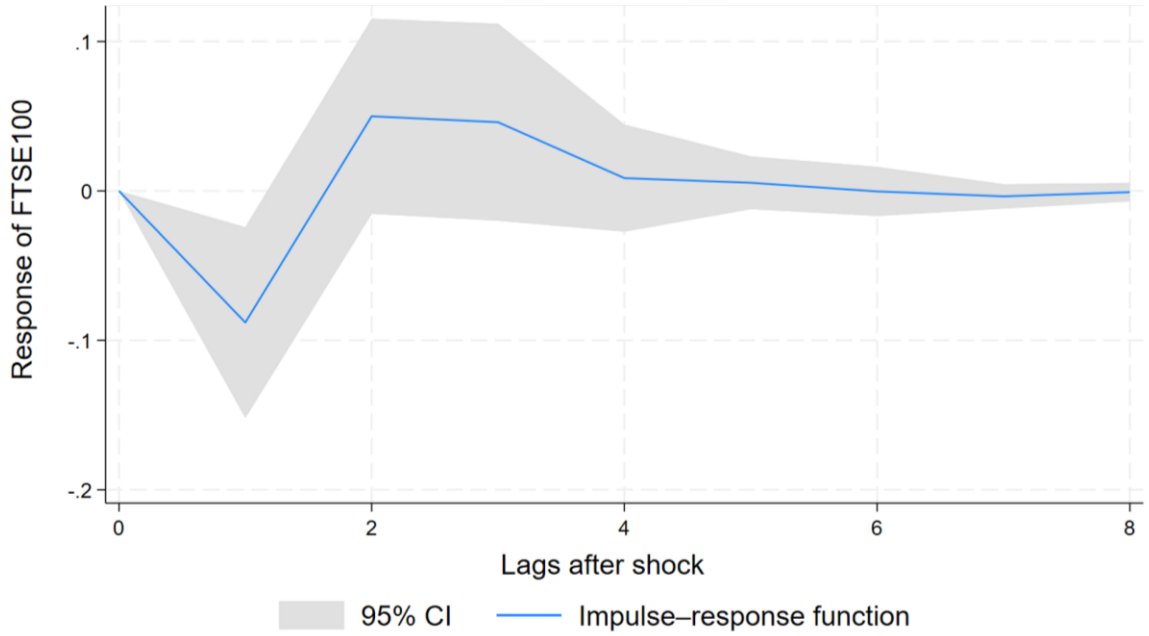
3.4 IRF (impulse response function)

Whilst impulse response analysis found no significance in the impact of variations in the Dollar/Sterling exchange rate on Brent crude oil prices, and vice versa, IRF analysis of the FTSE100 yielded notable results. The response of the FTSE100 to a one standard deviation shock in oil prices depicts a strong negative effect a single lag (one week) from the shock. This shock has a growing impact on the FTSE100 reaching a peak at week 4 before stabilising. This impact is not sustained as the market adjusts and the effects diminish, as illustrated by figure 2. On the other hand, the response of brent oil prices to a shock to the FTSE 100 seems immediate but diminishes steadily over time. This reflects the traditional economical outlook on the issue, that changes in oil prices have a constant and lasting effect on global economies whereas FTSE100 and other stock market fluctuations have a short-lived effect.

Table 6. *Impulse Response analysis of FTSE100 and Brent oil prices*

Lags	Response of FTSE100 to shock in Brent oil Prices	Response of Brent oil Prices to shock in FTSE100
1	0.022565	0.977435
2	0.061174	0.938826
3	0.074542	0.925458
4	0.084793	0.915207
5	0.084549	0.915451
6	0.08457	0.91543
7	0.084565	0.915435
8	0.084615	0.915385

Figure 2. *Response of FTSE100 to a One Standard Deviation shock in Brent crude oil prices*



3.5 Tests for known breaks

After determining the break points of interest, a chow test and a Wald test was run for each known structural break. The two variables (FTSE100 and the Dollar/Sterling exchange rate) were regressed on Brent crude oil prices and dummy variables representing the aforementioned breaks. As table 7 shows, “Break 3” did not yield positive p value results for either variable. Moreover, no p values for tests for structural breaks were significant for the model including the Dollar/Sterling exchange rate, reinforcing the relative significance of the FTSE100’s relationship with Brent crude oil prices.

Table 7. *Tests for structural breaks*

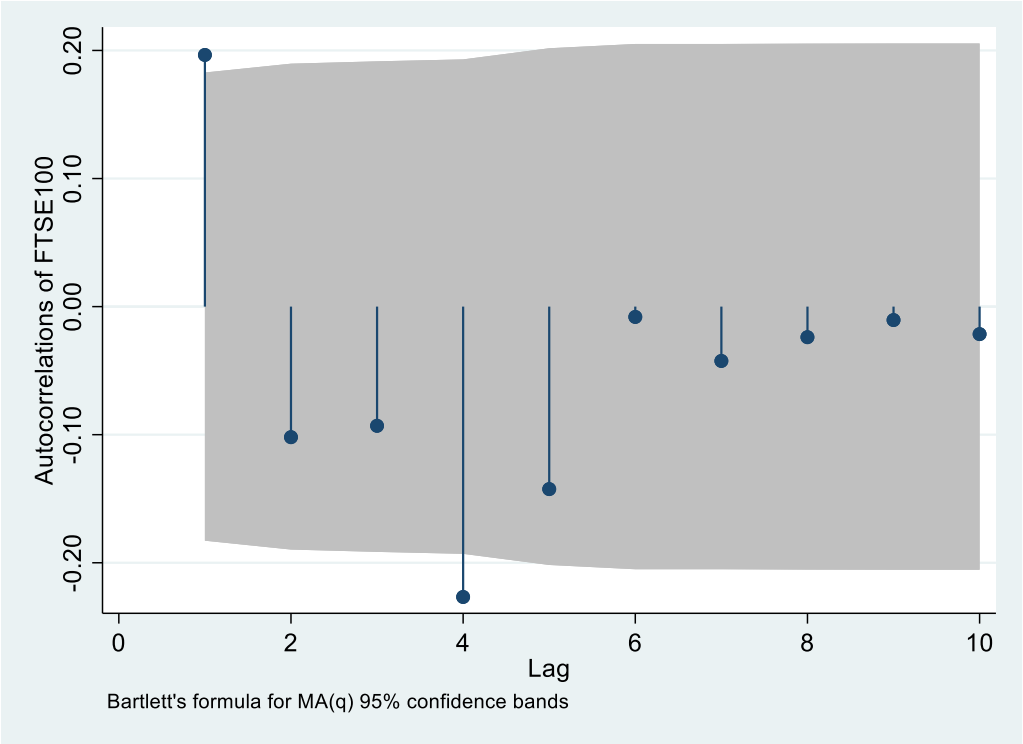
Break	F-Statistic for Model	p-value for F-Test	Wald Test Chi-Square	p-value for Wald Test	F-Statistic for Chow test	p-value for F-Test (Chow)
FTSE100						
Break 1	3.7000	0.0024**	5.9616	0.0508**	2.9800	0.0548*
Break 2	5.1000	0.0024**	0.9818	0.0068**	4.9900	0.0084**
Break 3	1.9100	0.1318	1.1650	0.5585	0.4100	0.6626
Dollar/Sterling Exchange Rate						
Break 1	1.2400	0.2983	3.0702	0.2154	1.5400	0.2200
Break 2	1.4600	0.2302	3.7145	0.1561	1.8600	0.1609
Break 3	0.6900	0.5592	1.0629	0.5877	0.0000	0.9966

Note: Results rounded to 4D.P

3.6.1 ARIMA modelling

Before incorporating the break variables into an ARIMA model, model specifications were tested comparatively, as shown in table 8. Individual ARIMA model AR and MA inputs were determined by partial-autocorrelation and autocorrelation plots respectively which, as shown in figures 3 and 4, highlights a significance in the first and fourth AR and MA terms.

Figure 3 and 4. *PACF and ACF correlograms, first 10 lags.*



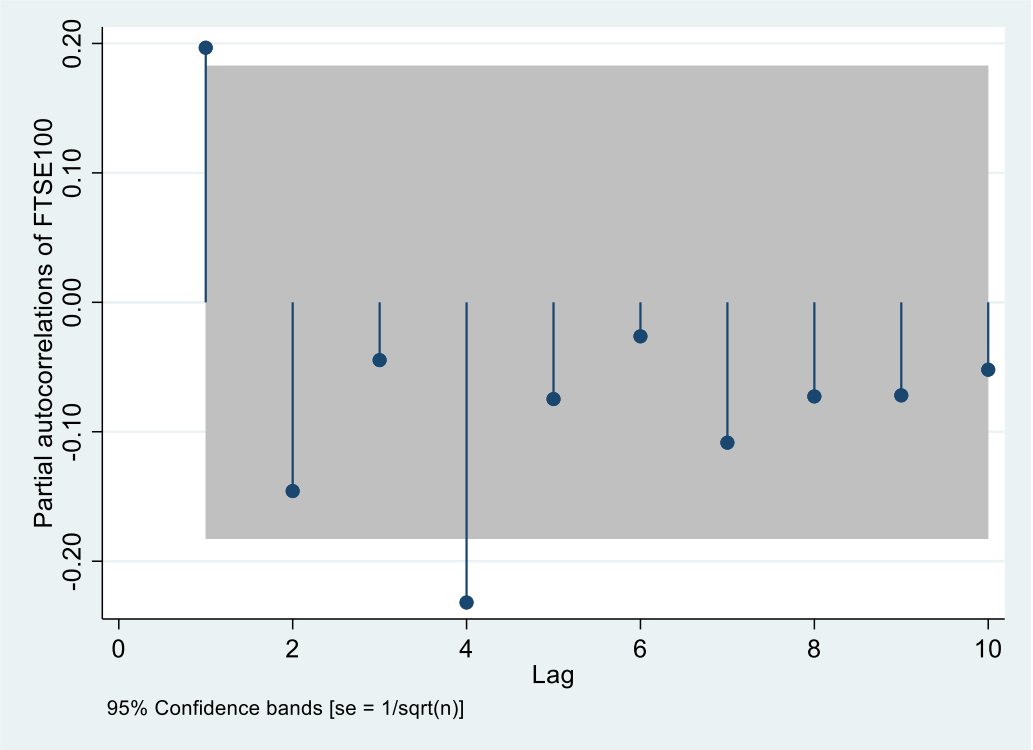


Table 8. *ARIMA model selection for FTSE100*

Model Selection Criteria	Model 1	Model 2	Model 3	Model 4	Best Model
ARMA Specification and significance	(1,1,1)	(4,1,4)	(4,1,1)	(1,1,4)	Model 3
Log likelihood	324.02	325.99	327.92	329.06	Model 4
AIC	-640.04	-648.50	-641.84	-644.13	Model 2
BIC	-629.06	-623.80	-622.62	-624.91	Model 3
Best Model	-	-	-	-	Model 3

Note: Results rounded to 2D.P

3.6.2 ARIMA modelling with breaks

By using the Akaike Information Criteria (AIC) and Bayesian Information Criteria to test model 3's fit and 'parsimony' when adding dummy break variables and the Brent crude oil variable, we can determine whether the breaks serve to strengthen the model. Table 9 reinforces the previous assessment of break 3, thereby concluding a lack of significance between the break dummy and the FTSE100 ARIMA model. This is truer given a 0.805 p-statistic for break 3 when included in the model in contrast to the <0.05 p-statistics for the other two breaks and Brent crude oil prices.

Table 9. *AIC and BIC tests of ARIMA 3 models*

Model Description	AIC	BIC
ARIMA model 3 (4,1,1)	-641.84	-622.62
Model 3 with Brent Oil	-643.41	-621.45
Model with Brent Oil + Breaks 1-3	-653.69	-626.24
Model with Brent Oil + Breaks 1-2	-654.44	-629.73

Note: Results rounded to 2D.P

4. Conclusion and discussion

Bidirectional Granger causality between the FTSE100 and Brent crude oil prices, along with the results from IRF analysis, points to the sensitivity of the UK economy to spikes in oil prices, calling for a vigilant and constant examination of the subject. Contrariwise, the effects of spikes in price of oil and a general relationship between Brent crude oil and the Dollar/Sterling exchange rate was less pronounced yet expected as a change in the eal price of the dollar would have an effect on Brent crude oil prices as barrels of oil are traded in dollars.

The statistical significance of two of three break points illustrates the danger posed by the global interconnectedness of markets, at the very least in the US and Europe. This piece stands testament against the growing sentiment of idealistic independence in the United Kingdom. Specifically, the Silicon Valley Bank crash's consequences echoed through the global market, having an effect on Brent oil prices and the UK by extension. Interestingly, however, break 3, in regard to an OPEC announcement, yielded no significant results perhaps alluding to the weakening influence of OPEC+ amidst diversification of global oil sources. As a culmination of this piece's findings, this analysis provides clarity to and highlights the UK's vulnerability to external shocks. Amid growing geopolitical tensions, economists and policymakers should emphasise caution and strategy as to accommodate for global market sensitivities and threats.

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Appendix:

Example of filling in missing data:

Ipolate dcoilbrenteu numdate, gen (dcoilbrenteu_i) epolate

Converting to weekly data:

Collapse (mean) (variables) by (weekly)

Setting weekly data for time series analysis:

Tsset numdate weekly

Smoothing data for graphical purposes:

Tssmooth ma MAsmootheddcoilbrenteu = dcoilbrenteu_i, window(2 1 2)

Generating log of variables:

Gen l_dcoilbrenteu=log(dcoilbrenteu_i)

Generating first differences of variable:

Gen d_l_dcoilbrenteu=D1.l_dcoilbrenteu

Testing for unknown break points:

Xtbreak d_l_dcoilbrenteu inter trend, vce(hac)

Augmented dickey fuller test:

Dfuller (variable)

Example for testing for lag selection for VAR:

Varsoc d_l_dcoilbrenteu d_l_ftse100

Example of VAR regression:

Var d_l_dcoilbrenteu d_l_ftse100, lag(1/3)

Testing for VAR stability conditions:

Varstable

Testing for Lagrange multiplier:

Varlmar

VAR granger-causality tests:

vargranger

Granger-causality tests with OLS:

Regress d_l_ftse100 L(1/4).d_l_ftse100 L(1/4).d_l_dcoilbrenteu

Test L1.d_l_dcoilbrenteu L2.d_l_dcoilbrenteu L3.d_l_dcoilbrenteu L4.d_l_dcoilbrenteu

IRF table:

Irf table fevd, irf(IRF) impulse(d_l_dcoilbrenteu d_l_ftse100) response(d_l_ftse100) noci

IRF graph:

irf graph irf, (impulse variable) (response variable)

Generating dummy break variable:

Gen break=1 if trend==67|trend>67

Chow test:

Reg d_l_ftse100 d_l_dcoilbrenteu break 1 break1oil

Test break1oil break1

Test for known structural break point:

Estat sbknown, break(tw(2022w49))

Partial autocorrelation/autocorrelation tests:

Pac d_l_ftse100

Ac d_l_ftse100

Corrgram d_l_ftse100

Constructing ARIMA model:

Arima d_l_dcoilbrenteu, arima (4,0,1)